Unified analysis of discontinuous Galerkin methods for elliptic problems: a Lean blueprint

Formalization of D. N. Arnold, F. Brezzi, B. Cockburn and L. D. Marini (2002)

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Chapter 1

Overview

This blueprint describes a complete Lean 4 formalization of the main results of

> D. N. Arnold, F. Brezzi, B. Cockburn, L. D. Marini, *Unified analysis of discontinuous Galerkin methods for elliptic problems*, SIAM J. Numer. Anal. **39** (2002), no. 5, 1749–1779.

The model problem is $-\Delta u = f$ in a bounded open convex polygon $\Omega \subset \mathbb{R}^2$, $u = 0$ on $\partial\Omega$. The paper writes discontinuous Galerkin (DG) methods in a unified flux formulation, eliminates the auxiliary variable to obtain primal forms, and analyses nine methods (Table 6.1 of the paper): the four consistent, adjoint-consistent and stable methods (interior penalty, Bassi et al., Brezzi et al., LDG), with optimal error estimates; ordinary NIPG and the two pure penalty methods; the superpenalized variants; the Baumann–Oden method; and the original Bassi–Rebay method, which is only weakly stable.

The Lean files.

- `Showcase.lean` imports only Mathlib. It defines the setting (weak derivatives, triangulations, broken fields with traces, liftings, the forms of all methods, norms and numerical fluxes) and states eight theorems in the paper's notation, with `sorry`.
- `Showcase_WithProofs.lean` proves the same theorems from the library `LeanCode`.
- `comparator.json` configures `lake comparator`. It checks that the proofs establish exactly the statements of `Showcase.lean` and use only the axioms `propext`, `Classical.choice` and `Quot.sound`.
- The library `LeanCode` contains about 31 000 lines of proofs, organized as in the chapters below. `LeanCode/Definitions.lean` repeats the definitions of `Showcase.lean` verbatim; `LeanCode/Main/Bundled.lean` states the results in bundled form and `LeanCode/Main/Proofs.lean` derives the eight theorems from them.
- `blueprint/DetailedProofs.pdf` contains detailed mathematical proofs of these results. In places the Lean proofs take a different route, notably for the H^2 regularity on convex polygons, which is proved here without the Fourier transform.

Scope. Every theorem holds for every polynomial degree $p \geq 1$ (for Baumann–Oden $p \geq 2$, with a counterexample for $p = 1$) and every shape bound κ; meshes are conforming triangulations of Ω with $h_K/\rho_K \leq \kappa$, without quasi-uniformity. All constants are chosen before the mesh, the edge coefficients, the load and the exact solution.

Chapter 2

Results from Mathlib

The formalization builds on Mathlib. The following definitions and results of Mathlib are its main ingredients; in the dependency graph they are grey, and their edges show where they enter the proofs.

Definition 2.1 (The Hilbert space L^2). [Lean: `MeasureTheory.Lp`, `MeasureTheory.L2.innerProductSpace`, `MeasureTheory.Lp.instCompleteSpace`] For a measure μ, the space $L^2(\mu)$ of square-integrable functions, taken modulo equality almost everywhere, with its inner product; it is complete.

Theorem 2.2 (Riesz representation and Lax–Milgram theorems). [Lean: `InnerProductSpace.toDual`, `IsCoercive.continuousLinearEquivOfBilin`] *Every bounded linear functional on a Hilbert space is of the form $x \mapsto (y, x)$, and a bounded coercive bilinear form B induces an isomorphism: for every bounded functional ℓ there is a unique u with $B(u, v) = \ell(v)$ for all v.*

Definition 2.3 (Affine simplices). [Lean: `Affine.Simplex`, `Affine.Simplex.closedInterior`, `Affine.Simplex.faceOpposite`] Affine simplices in a Euclidean space, their closed interiors and their faces.

Definition 2.4 (Hausdorff measure). [Lean: `MeasureTheory.Measure.euclideanHausdorffMeasure`] The k-dimensional Hausdorff measure on a Euclidean space, normalized to agree with Lebesgue measure on k-dimensional subspaces; it is the measure on the faces of the cells.

Definition 2.5 (Multivariate polynomials). [Lean: `MvPolynomial`, `MvPolynomial.pderiv`] Polynomials in d variables with real coefficients, and their partial derivatives.

Theorem 2.6 (Fubini–Tonelli theorem). [Lean: `MeasureTheory.integral_prod`, `MeasureTheory.lintegral_prod`, `MeasureTheory.integral_integral_swap`] *Integrals over a product of σ-finite measure spaces are iterated integrals, in either order.*

Theorem 2.7 (Fundamental theorem of calculus). [Lean: `intervalIntegral.integral_eq_sub_of_hasDerivAt`] $\int_a^b f' = f(b) - f(a)$ *when f has an integrable derivative on $[a, b]$.*

Theorem 2.8 (Bump functions and convolution). [Lean: `ContDiffBump`, `HasCompactSupport.contDiff_convolution_left`] *There are smooth bump functions with compact support, and the convolution of a locally integrable function with a smooth compactly supported kernel is smooth.*

Theorem 2.9 (Fundamental lemma of the calculus of variations). [Lean: `IsOpen.ae_eq_zero_of_integral_contDiff_smul_eq_zero`] *A locally integrable function on an open set whose integrals against all smooth compactly supported functions vanish is zero almost everywhere.*

Theorem 2.10 (Linear maps of finite-dimensional spaces). [Lean: `LinearMap.injective_iff_surjective`] *A linear endomorphism of a finite-dimensional vector space is injective if and only if it is surjective.*

Theorem 2.11 (Change of variables). [Lean: `MeasureTheory.integral_image_eq_integral_abs_det_fderiv_smul`] $\int_{\Phi(S)} f = \int_S |\det D\Phi|\,(f \circ \Phi)$ *for injective maps Φ that are differentiable on S.*

Chapter 3

Setting and definitions

All definitions are in `Showcase.lean`, namespace `ABCMShowcase`. They are repeated verbatim in `LeanCode/Definitions.lean`.

Definition 3.1 (Functions, jets and Sobolev norms)**.** [Lean: `ABCMShowcase.Space, ABCMShowcase.Index, ABCMShowcase.Jet, ABCMShowcase.order, ABCMShowcase.smoothJet, ABCMShowcase.Smooth, ABCMShowcase.WeakJet, ABCMShowcase.squareL2, ABCMShowcase.seminormSq, ABCMShowcase.sobolevNormSq, ABCMShowcase.gradient, ABCMShowcase.laplacian`] A *jet* on $\mathbb{R}^d$ assigns a function J_α to every multi-index α; the jet of a smooth function u is $\alpha \mapsto \partial^\alpha u$. A jet is a *weak jet of order* m on an open set U if $J_\alpha \in L^2(U)$ for $|\alpha| \le m$ and $J_{\alpha+e_i}$ is the weak derivative $\partial_i J_\alpha$ for $|\alpha| < m$. The seminorm $|J|_{H^m}$ sums the squared L^2 norms over all ordered derivative words of length m (for $m = 2$ the Frobenius norm of the Hessian), and $\|J\|_{H^m}^2 = \sum_{j \le m} |J|_{H^j}^2$.

Definition 3.2 (The domain and the Dirichlet problem)**.** [Lean: `ABCMShowcase.ConvexPolygon, ABCMShowcase.InH01, ABCMShowcase.InH2H01, ABCMShowcase.IsWeakSolution, ABCMShowcase.IsL2, ABCMShowcase.InH, ABCMShowcase.normL2, ABCMShowcase.seminormH, ABCMShowcase.normH`] A *convex polygon* is a nonempty bounded finite intersection of open half-planes. A jet is in $H_0^1(\Omega)$ if it is a weak jet of order one and an H^1-limit of smooth functions with compact support in Ω. A *weak solution* of $-\Delta u = f$, $u = 0$ on $\partial\Omega$, is a jet $u \in H_0^1(\Omega)$ with $(\nabla u, \nabla v)_\Omega = (f, v)_\Omega$ for all smooth compactly supported v.

Definition 3.3 (Triangulations)**.** [Lean: `ABCMShowcase.MeshData, ABCMShowcase.Cells, ABCMShowcase.Faces, ABCMShowcase.cellSet, ABCMShowcase.facet, ABCMShowcase.faceSet, ABCMShowcase.cellSize, ABCMShowcase.AdmissibleMesh, ABCMShowcase.meshSize, ABCMShowcase.faceLength`] A mesh consists of finitely many affine triangles K, a list of edges e with an owner side and an optional neighbour side, and outward unit normals. It is *admissible* for Ω with shape bound κ if the closed triangles cover $\overline{\Omega}$, have disjoint interiors and meet in common sub-simplices, $h_K \le \kappa \rho_K$ for an inscribed ball of radius ρ_K, and the edge list matches the triangle sides, exactly the edges without neighbour lying on $\partial\Omega$. $h = \max_K h_K$ and h_e is the length of e.

Definition 3.4 (Broken fields, traces and the discrete spaces)**.** [Lean: `ABCMShowcase.BrokenField, ABCMShowcase.HasSobolevTrace, ABCMShowcase.BrokenH2, ABCMShowcase.RealizesGlobalJet, ABCMShowcase.TraceRealization, ABCMShowcase.PolyField, ABCMShowcase.polynomialField, ABCMShowcase.DegreeBound, ABCMShowcase.InWh, ABCMShowcase.subField`] A *broken field* has a jet on every triangle and a trace jet on every side. The traces of a jet on a triangle are *Sobolev traces* if they are L^2 limits of the restrictions of smooth functions whose derivatives converge in L^2 of the triangle; a broken field is *broken* H^2 if every cell jet is an H^2 jet with such traces. A broken field *realizes* a global jet u if its cell jets are those of u. V_h consists of the piecewise polynomials of degree at most p, and $W_h = V_h + H^2(\Omega) \cap H_0^1(\Omega)$ of the broken fields that are sums of a discrete field and a field realizing an $H^2 \cap H_0^1$ jet.

Definition 3.5 (Averages, jumps and discrete vector fields)**.** [Lean: `ABCMShowcase.jump, ABCMShowcase.avg, ABCMShowcase.avgGrad, ABCMShowcase.jumpGrad, ABCMShowcase.VecPolyField, ABCMShowcase.`

VecDegreeBound, ABCMShowcase.vecInner, ABCMShowcase.vecAvg, ABCMShowcase.vecJump] On an interior edge, $[\![v]\!] = v^+n^+ + v^-n^-$, $\{v\} = (v^+ + v^-)/2$, $\{\nabla v\} = (\nabla v^+ + \nabla v^-)/2$ and $[\![\nabla v]\!] = \nabla v^+ \cdot n^+ + \nabla v^- \cdot n^-$; on a boundary edge $[\![v]\!] = vn$ and $\{\nabla v\} = \nabla v$. $\Sigma_h = V_h^2$ is the space of piecewise polynomial vector fields, with the $L^2(\Omega)^2$ inner product and the edge average and jump of vector fields.

Definition 3.6 (Liftings). [Lean: ABCMShowcase.IsLiftR, ABCMShowcase.IsLiftL, ABCMShowcase.liftR, ABCMShowcase.liftL, ABCMShowcase.liftRSum, ABCMShowcase.liftLSum, ABCMShowcase.liftA] For an edge e and $\varphi \in L^2(e)^2$, $r_e(\varphi) \in \Sigma_h$ is defined by $(r_e(\varphi), \tau)_\Omega = -\int_e \varphi \cdot \{\tau\}$ for all $\tau \in \Sigma_h$; for an interior edge and $q \in L^2(e)$, $l_e(q) \in \Sigma_h$ is defined by $(l_e(q), \tau)_\Omega = -\int_e q[\![\tau]\!]$. Then $R(v) = \sum_e r_e([\![v]\!])$, $L_\beta(v) = \sum_{e \text{ interior}} l_e(\beta_e \cdot [\![v]\!])$ and $A_\beta = R + L_\beta$. In Lean the liftings are defined by choice from their defining properties; existence and uniqueness are proved in Lemma 8.1.

Definition 3.7 (The bilinear forms). [Lean: ABCMShowcase.cellGrad, ABCMShowcase.fluxC, ABCMShowcase.fluxD, ABCMShowcase.penaltyJ, ABCMShowcase.penaltyR, ABCMShowcase.superJ, ABCMShowcase.superR, ABCMShowcase.formIP, ABCMShowcase.formBassi, ABCMShowcase.formBrezzi, ABCMShowcase.formLDG, ABCMShowcase.formNIPG, ABCMShowcase.formPenaltyJ, ABCMShowcase.formPenaltyR, ABCMShowcase.formBO, ABCMShowcase.formBR, ABCMShowcase.formSuperJ, ABCMShowcase.formSuperR, ABCMShowcase.formSuperNIPG, ABCMShowcase.StdMethod, ABCMShowcase.StdMethod.form, ABCMShowcase.StdMethod.Threshold] With $g(w,v) = (\nabla_h w, \nabla_h v)_\Omega$, $C(w,v) = \sum_e \langle \{\nabla_h w\}, [\![v]\!] \rangle_e$, $C^T(w,v) = C(v,w)$, $D_\beta(w,v) = \sum_{e \text{ interior}} \langle [\![\nabla_h w]\!], \beta_e \cdot [\![v]\!] \rangle_e$, $A_j(w,v) = \sum_e \eta_e h_e^{-1} \langle [\![w]\!], [\![v]\!] \rangle_e$, $A_r(w,v) = \sum_e \eta_e (r_e([\![w]\!]), r_e([\![v]\!]))_\Omega$ and the superpenalties $A_j^{\sup}$, $A_r^{\sup}$ (weights $h_e^{-(2p+1)}$ and h_e^{-2p}):

$$
\begin{aligned}
B_{IP} &= g - C - C^T + A_j, \qquad B_{Bassi} = g - C - C^T + A_r, \\
B_{Brezzi} &= g - C - C^T + (R(\cdot), R(\cdot)) + A_r, \\
B_{LDG} &= g - C - C^T - D_\beta - D_\beta^T + (A_\beta(\cdot), A_\beta(\cdot)) + A_j, \\
B_{NIPG} &= g - C + C^T + A_j, \qquad B_{BO} = g - C + C^T, \qquad B_{BR} = g - C - C^T + (R(\cdot), R(\cdot)),
\end{aligned}
$$

the pure penalties $g + A_j$ and $g + A_r$, and the superpenalized $g + A_j^{\sup}$, $g + A_r^{\sup}$, $g - C + C^T + A_j^{\sup}$. The penalty threshold of the four standard methods is $\eta_{\min} > \eta_*$ for IP, $\eta_{\min} > 3$ for Bassi et al., and none for Brezzi et al. and LDG.

Definition 3.8 (Norms). [Lean: ABCMShowcase.gradSq, ABCMShowcase.hessSq, ABCMShowcase.jumpSq, ABCMShowcase.normN, ABCMShowcase.superNormJ, ABCMShowcase.superNormR, ABCMShowcase.l2Norm] $N_h(v)^2 = G_h(v)^2 + H_h(v)^2 + J_h(v)^2$ with $G_h(v)^2 = \sum_K \|\nabla v\|_K^2$, $H_h(v)^2 = \sum_K h_K^2 |v|_{H^2(K)}^2$ and $J_h(v)^2 = \sum_e h_e^{-1} \|[\![v]\!]\|_e^2$. The superpenalty norms replace J_h^2 by $A_j^{\sup}(v,v)$ or $A_r^{\sup}(v,v)$.

Definition 3.9 (Discrete problems). [Lean: ABCMShowcase.loadPair, ABCMShowcase.fieldPair, ABCMShowcase.IsDiscreteSolution] *u_h solves the discrete problem* for a form B and a load f if $u_h \in V_h$ and $B(u_h, v_h) = (f, v_h)_\Omega$ for all $v_h \in V_h$.

Definition 3.10 (Discrete gradient, projection and the Bassi–Rebay kernel). [Lean: ABCMShowcase.IsCellProj, ABCMShowcase.cellProj, ABCMShowcase.discreteGrad, ABCMShowcase.InBRKernel, ABCMShowcase.PiecewisePoly] P_k is the elementwise L^2 projection onto polynomials of degree at most k; $\mathcal{D}_h v_h = \nabla_h v_h + R(v_h) \in \Sigma_h$ is the discrete gradient; $Z_h = \{v_h \in V_h : \mathcal{D}_h v_h = 0\}$ is the Bassi–Rebay kernel.

Definition 3.11 (Numerical fluxes and the flux formulation). [Lean: ABCMShowcase.Method, ABCMShowcase.Method.form, ABCMShowcase.FluxTable, ABCMShowcase.Method.fluxes, ABCMShowcase.Method.auxField, ABCMShowcase.IsFluxSolution, ABCMShowcase.outFlux] For each of the nine methods and the three superpenalized methods, the numerical fluxes $\hat{u}$, $\hat{\sigma}$ of Table 3.1 of the paper. A pair $(u_h, \sigma_h) \in V_h \times \Sigma_h$ solves the *flux formulation* if on every triangle K

$$
(\sigma_h, \tau)_K = -(u_h, \operatorname{div} \tau)_K + \langle \hat{u}, \tau \cdot n_K \rangle_{\partial K}, \qquad (\sigma_h, \nabla v)_K = (f, v)_K + \langle \hat{\sigma} \cdot n_K, v \rangle_{\partial K}
$$

for all polynomial τ, v of degree at most p. The eliminated auxiliary field is $\nabla_h u_h + R(u_h)$, $\nabla_h u_h + A_\beta(u_h)$, $\nabla_h u_h - R(u_h)$ or $\nabla_h u_h$, depending on the method.

Chapter 4

The main theorems

Each theorem is stated in `Showcase.lean`, as a proposition in `LeanCode/Statements.lean`, proved in `LeanCode/Main/Proofs.lean` from a bundled result of `LeanCode/Main/Bundled.lean`, and restated in `Showcase_WithProofs.lean`.

Theorem 4.1 (The Dirichlet problem on a convex polygon)**.** [Lean: `ABCMShowcase.dirichlet_problem`, `ABCMShowcase.DirichletProblem`, `ABCM.dirichletProblem`, `ABCM.PoissonResults`, `ABCM.poissonResults`] *Let Ω be a bounded open convex polygon. There is $C_{reg} > 0$ such that for every $f \in L^2(\Omega)$ the problem $-\Delta u = f$, $u = 0$ on $\partial\Omega$ has a weak solution $u \in H^1_0(\Omega)$, unique up to its derivatives of order at most one, and $u \in H^2(\Omega)$ with $-\Delta u = f$ and $\|u\|_{H^2(\Omega)} \le C_{reg}\|f\|_{L^2(\Omega)}$. Every $H^2(\Omega)$ jet has Sobolev traces on the edges of every admissible mesh.*

Proof. Existence and uniqueness by Lax–Milgram (Lemma 6.1); the Hessian and its bound by Theorem 6.8 and the energy estimate; the traces by Lemma 5.4. □

Theorem 4.2 (Section 5.1: the four stable methods)**.** [Lean: `ABCMShowcase.four_methods_theorem`, `ABCMShowcase.FourMethodsTheorem`, `ABCM.fourMethodsTheorem`, `ABCM.StandardFourResults`, `ABCM.standardFourResults`] *Fix $p \ge 1$ and $\kappa > 0$. There is $\eta_* = \eta_*(p,\kappa) > 0$ such that for every convex polygon Ω, all $0 < \eta_{\min} \le \eta_{\max}$, $b \ge 0$ and every method among IP, Bassi et al., Brezzi et al. and LDG satisfying its threshold, there are $C_b, C_s, C_E, C_0 > 0$ with the following properties for every admissible mesh, all $\eta_e \in [\eta_{\min}, \eta_{\max}]$ and all $|\beta_e| \le b$:*

1. *B_h is symmetric and bilinear on W_h, $|B_h(w,v)| \le C_b N_h(w) N_h(v)$, and $B_h(v_h, v_h) \ge C_s N_h(v_h)^2$ on V_h;*
2. *for every $f \in L^2(\Omega)$ the discrete problem has exactly one solution u_h;*
3. *if the weak solution u lies in $H^{p+1}(\Omega)$, then $N_h(u - u_h) \le C_E h^p |u|_{H^{p+1}(\Omega)}$ and $\|u - u_h\|_{L^2(\Omega)} \le C_0 h^{p+1} |u|_{H^{p+1}(\Omega)}$.*

Proof. Boundedness and coercivity are Lemmas 8.3 and 8.4; solvability follows by Lemma 8.5. With consistency (Lemma 8.7), the abstract energy estimate (Lemma 8.8) and the interpolant of Lemma 7.4 give the energy bound. The L^2 bound follows by duality (Lemma 8.9), using adjoint consistency and the H^2 regularity of Theorem 4.1. □

Theorem 4.3 (Section 3.3: consistency, adjoint consistency, Galerkin orthogonality)**.** [Lean: `ABCMShowcase.consistency_theorem`, `ABCMShowcase.ConsistencyTheorem`, `ABCM.consistencyTheorem`, `ABCM.StandardConsistency`, `ABCM.standardConsistency`, `ABCM.StandardOrthogonality`, `ABCM.standardOrthogonality`] *For each of the four stable methods, with arbitrary η_e and β_e: if $u \in H^2(\Omega) \cap H^1_0(\Omega)$ and $-\Delta u = f$, then $B_h(u,v) = B_h(v,u) = (f,v)_\Omega$ for all $v \in W_h$, and every discrete solution satisfies $B_h(u - u_h, v_h) = 0$ for all $v_h \in V_h$.*

Proof. Lemma 8.7; Galerkin orthogonality follows by linearity. □

Theorem 4.4 (Section 3.2: flux and primal formulations). [Lean: `ABCMShowcase.flux_formulations`, `ABCMShowcase.FluxFormulations`, `ABCM.fluxFormulations`, `ABCM.FluxCorrespondence`, `ABCM.fluxCorrespondence`] *For each of the twelve methods and arbitrary η_e, β_e, a pair (u_h, σ_h) solves the flux formulation if and only if σ_h is the eliminated auxiliary field of u_h and u_h solves the primal problem $B_h(u_h, v_h) = (f, v_h)_\Omega$. Every flux solution satisfies the balance $\int_S f + \int_{\partial S} \hat{\sigma} \cdot n_S = 0$ for every set S of triangles.*

Proof. The first equation of the flux formulation determines σ_h (Lemma 9.1); substituting it into the second gives the primal form. For the balance, sum the second equation with $v = 1$ over S; the contributions of interior faces cancel. □

Theorem 4.5 (Section 5: NIPG, pure penalties and superpenalties). [Lean: `ABCMShowcase.penalty_methods_theorem`, `ABCMShowcase.PenaltyMethodsTheorem`, `ABCM.penaltyMethodsTheorem`, `ABCM.PenaltyFamilyResults`, `ABCM.penaltyFamilyResults`] *Fix $p \geq 1$, κ, Ω and $0 < \eta_{\min} \leq \eta_{\max}$. With constants uniform over admissible meshes and $\eta_e \in [\eta_{\min}, \eta_{\max}]$:*

- *NIPG is bounded on W_h, coercive on V_h and uniquely solvable, and for $u \in H^{p+1}(\Omega)$: $N_h(u - u_h) \leq C_E h^p |u|_{H^{p+1}}$ and $\|u - u_h\|_{L^2} \leq C_L h^p |u|_{H^{p+1}}$;*
- *the pure penalty methods $g + A_j$ and $g + A_r$ are bounded, coercive and uniquely solvable;*
- *the three superpenalized methods are uniquely solvable, and their errors satisfy h^p bounds in the superpenalty norms and h^{p+1} bounds in L^2, in terms of $\|u\|_{H^{p+1}}$ (of $|u|_{H^{p+1}}$ for superpenalized NIPG).*

Proof. Stability is Lemma 10.1, solvability Lemma 8.5. For NIPG the energy estimate follows from consistency (Theorem 4.6) and Lemma 8.8; the L^2 bound from the broken Poincaré inequality (Lemma 10.2). For the superpenalties the consistency defect $C(u, \cdot)$ is of order h^p in the superpenalty norm by the jump control of Lemma 10.3, and Lemma 10.4 gives both estimates. □

Theorem 4.6 (Table 6.1: consistency defects). [Lean: `ABCMShowcase.penalty_consistency_defects`, `ABCMShowcase.PenaltyConsistencyDefects`, `ABCM.penaltyConsistencyDefects`, `ABCM.PenaltyConsistency`, `ABCM.penaltyConsistency`] *Let $u \in H^2(\Omega) \cap H^1_0(\Omega)$ with $-\Delta u = f$. For all $v \in W_h$ and arbitrary η_e: NIPG and superpenalized NIPG satisfy $B(u, v) = (f, v)_\Omega$ and $B(v, u) = (f, v)_\Omega + 2C(u, v)$; the pure penalties and the pure superpenalties satisfy $B(u, v) = B(v, u) = (f, v)_\Omega + C(u, v)$.*

Proof. The identity $g(u, v) - C(u, v) = (f, v)_\Omega$ (Lemma 8.6) and the vanishing of the jumps of u, which removes all penalty terms and $C(v, u)$. □

Theorem 4.7 (Section 5.3.1: the Baumann–Oden method). [Lean: `ABCMShowcase.baumann_oden_theorem`, `ABCMShowcase.BaumannOdenTheorem`, `ABCM.baumannOdenTheorem`, `ABCM.BaumannOdenResults`, `ABCM.baumannOdenResults`, `ABCM.BaumannOdenDegreeOne`, `ABCM.baumannOdenDegreeOne`] *For $p \geq 2$, every admissible mesh and every $f \in L^2(\Omega)$ the Baumann–Oden problem has exactly one solution, and if $u \in H^{p+1}(\Omega)$, then $N_h(u - u_h) \leq C_E h^p |u|_{H^{p+1}}$ and $\|u - u_h\|_{L^2} \leq C_L h^p |u|_{H^{p+1}}$, with uniform constants. For $p = 1$ there is a one-triangle mesh on which $B_{BO}(w, v) = (\nabla w, \nabla v)_\Omega$ for affine w, v, and on which the load $f = 1$ has no discrete solution.*

Proof. Stability and solvability are Lemma 11.1; the energy estimate uses the consistency of B_{BO} and a test with piecewise quadratics that control the jumps of piecewise constants. The counterexample is Lemma 11.2. □

Theorem 4.8 (Section 5.3.2: the original Bassi–Rebay method). [Lean: `ABCMShowcase.bassi_rebay_theorem`, `ABCMShowcase.BassiRebayTheorem`, `ABCM.bassiRebayTheorem`, `ABCM.BassiRebayResults`, `ABCM.bassiRebayResults`, `ABCM.BassiRebayPolyLoad`, `ABCM.bassiRebayPolyLoad`] *For $p \geq 1$:*

- *$Z_h = \{v_h \in V_h : P_{p-1} v_h = 0$ on every triangle, $\{v_h\} = 0$ on every interior edge$\}$;*

- *the problem* $B_{BR}(u_h, v_h) = (f, v_h)_\Omega$ *is solvable if and only if* $f \perp Z_h$*; its solutions form one coset* $u_h + Z_h$*, and* $\mathcal{D}_h u_h$ *and* $P_{p-1}u_h$ *are unique;*
- *it is solvable for elementwise polynomial loads of degree at most* $p-1$*, and then, for* $u \in H^{p+1}(\Omega)$*,* $\|\nabla u - \mathcal{D}_h u_h\| \le C_\sigma h^p |u|_{H^{p+1}}$ *and* $\|P_{p-1}(u - u_h)\| \le C_0 h^{p+1}|u|_{H^{p+1}}$*;*
- *the full* L^2 *error is not controlled: on a one-triangle mesh there is* $0 \neq z \in Z_h$*, every* tz *solves the problem with* $f = 0$*, and* $\|0 - tz\| = |t|\|z\|$*.*

Proof. Lemmas 12.3, 12.4, 12.6 and 12.7. □

Chapter 5

Analysis on triangles

Lemma 5.1 (Geometry of admissible meshes). [Lean: `ABCM.MeshOK`, `ABCM.normals_eq_outwardNormal`, `ABCM.cellFace`, `ABCM.two_mul_le_height`, `ABCM.height_le_cellSize`] *An admissible mesh is a conforming shape-regular simplicial mesh; its stored normals are the unit outward normals of the triangles, every side of a triangle belongs to a unique edge, and every height H_j of a triangle satisfies $2\rho_K \le H_j \le h_K$.*

Proof. Elementary simplex geometry, using the affine-simplex library of Mathlib. □

Lemma 5.2 (Divergence theorem and cone formula on a simplex). [Lean: `ABCM.integral_fderiv_eq_sum_facets`, `ABCM.integral_closedInterior_cone`] *For a full-dimensional simplex s with facets F_j and unit outward normals ν_j, every C^1 function g and every direction w: $\int_s \nabla g \cdot w = \sum_j (w \cdot \nu_j) \int_{F_j} g$. Integrals over s can be computed as cone integrals over a facet.*

Proof. The cone formula $\int_s f = H_i \int_0^1 t^n \int_{F_i} f(v_i + t(y - v_i))\,\mathrm{d}y\,\mathrm{d}t$ reduces the radial derivative to facet integrals; combining the radial identities at two vertices gives every direction, since the edge vectors span the space. □

Lemma 5.3 (Density of smooth functions). [Lean: `ABCM.exists_smooth_approx`] *On an open convex set containing a ball, every weak jet of order m is the H^m limit of the jets of smooth functions.*

Proof. Dilate towards the centre of the ball, which maps the closure into the set, and mollify. □

Lemma 5.4 (Sobolev traces). [Lean: `ABCM.smooth_trace_ineq`, `ABCM.hasSobolevTrace_trace_ineq`, `ABCM.exists_hasSobolevTrace`, `ABCM.traceRealizationExists`] *For C^1 functions on a shape-regular cell, $h_K \|w\|^2_{L^2(F_j)} \le \frac{\rho}{2}\big((d+1)\|w\|^2_{L^2(K)} + h_K^2 \|\nabla w\|^2_{L^2(K)}\big)$. Every weak jet of order m on a cell has Sobolev traces, and every jet of order $k \ge 1$ on the domain has a broken realization with Sobolev traces on every admissible mesh.*

Proof. The trace inequality follows from the radial identity of Lemma 5.2. For the traces, approximate by smooth functions (Lemma 5.3); the trace inequality makes the facet restrictions Cauchy in L^2. □

Lemma 5.5 (Green's formula for jets with traces). [Lean: `ABCM.hasSobolevTrace_green`] *If J has Sobolev traces t of order m on a cell T and $|\alpha| < m$, then for every C^1 function v and every direction a: $\int_T (a \cdot \nabla J_\alpha) v + \int_T J_\alpha\, a \cdot \nabla v = \sum_j (a \cdot n_j) \int_{F_j} t_j^\alpha v$.*

Proof. Lemma 5.2 for smooth approximants and passage to the limit. □

Lemma 5.6 (Traces across interior edges). [Lean: `ABCM.traceAgreement`, `ABCM.jump_eq_zero_of_realizes`, `ABCM.trace_eq_zero_of_inH01`] *If J is a weak jet of order m on the domain with Sobolev traces on every cell, then for $|\alpha| < m$ the traces from the two sides of an interior edge agree almost everywhere. In particular $H^2 \cap H^1_0$ fields have zero jumps on all edges, and traces of H^1_0 jets vanish on boundary edges.*

Proof. Green's formula on both cells and the weak derivative on the domain give $\int_e (t_T - t_S)\psi = 0$ for all smooth ψ supported near the open edge; the fundamental lemma of the calculus of variations on the edge gives $t_T = t_S$ almost everywhere. □

Lemma 5.7 (Inverse and discrete trace inequalities). [Lean: `ABCM.exists_cell_inverse, ABCM.exists_cell_inverse_H2, ABCM.exists_cell_trace`] *For polynomials q of degree at most k on the cells of admissible meshes with shape bound ρ: $h_K^2 \|\partial_i q\|_K^2 \le C\|q\|_K^2$ (and the analogous H^2–H^1 bound), and $h_K \|q\|_{F_j}^2 \le C\|q\|_K^2$, with $C = C(d, k, \rho)$.*

Proof. Equivalence of norms on polynomials of bounded degree on the unit ball, transported by the affine maps between an inscribed and a circumscribed ball; the trace bound uses the radial identity. □

Lemma 5.8 (Poincaré–Wirtinger on convex sets). [Lean: `ABCM.poincare_convex`] *For a compact convex set K of diameter at most h and $u \in C^1$: $|K| \int_K u^2 - (\int_K u)^2 \le 2^{d+1} h^2 |K| \int_K |\nabla u|^2$.*

Proof. Split $u(x) - u(y)$ at the midpoint of $[x, y]$ and write each half as a segment integral of ∇u along a dilation towards one end point. □

Lemma 5.9 (Bramble–Hilbert). [Lean: `ABCM.exists_poly_moments, ABCM.exists_brambleHilbert`] *For a weak jet J of order $k+1$ on a cell T of an admissible mesh there is a polynomial q of degree at most k with $h_T^{2j} |J - q|_{H^j(T)}^2 \le C h_T^{2(k+1)} |J|_{H^{k+1}(T)}^2$ for all $j \le k+1$, with $C = C(d, k, \rho)$.*

Proof. Choose q with $\int_T \partial^\alpha (J - q) = 0$ for $|\alpha| \le k$ (polynomials with prescribed moments of all derivatives) and apply Lemma 5.8 repeatedly, first to smooth approximants. □

Lemma 5.10 (Scaled Sobolev sup bound). [Lean: `ABCM.exists_sobolev_sup`] *For $d \le 3$, smooth u and every point x of a cell T: $|T| u(x)^2 \le C\big(\|u\|_T^2 + h_T^2 |u|_{H^1(T)}^2 + h_T^4 |u|_{H^2(T)}^2\big)$, with $C = C(d)$.*

Proof. Taylor's formula along segments from x, averaged over $y \in T$, with the remainder controlled in L^2; the restriction $d \le 3$ makes the resulting weight integrable. □

Chapter 6

The Dirichlet problem on a convex polygon

Lemma 6.1 (Weak solutions). [Lean: `ABCM.laxMilgram_existsUnique, ABCM.poincare_H01, ABCM.existsUnique_weakSolution`] *A coercive bounded bilinear form on a Hilbert space has a unique solution for every bounded functional. If U lies in a slab of width L, then $\|v\| \le L\|\partial_{i_0} v\|$ on $H^1_0(U)$. Consequently, on a bounded set, $-\Delta u = f$ has a weak solution in H^1_0, unique up to its derivatives of order at most one.*

Proof. $H^1_0(U)$ is the closure of the graphs of test functions in $L^2(U)^{1+d}$, hence complete; the Poincaré inequality gives coercivity of $(\nabla u, \nabla v)$. □

Lemma 6.2 (Regularity on translation-invariant domains). [Lean: `ABCM.exists_dir_deriv, ABCM.exists_hessian_transInv`] *For a coercive constant-coefficient problem on a domain invariant under translations along a, the solution has weak derivatives along a in H^1_0, bounded by the load. In particular, on the plane and on half-planes, $v \in H^1_0$ with $(\nabla v, \nabla \varphi) = (F, \varphi)$ has weak second derivatives with $\|\partial_{ij} v\| \le 11\|F\| + 10\|v\|_{H^1}$.*

Proof. Difference quotients along a preserve H^1_0 and solve the same problem with a differenced load; they are bounded and converge. The normal second derivative is recovered from the equation. □

Lemma 6.3 (Sharp Wirtinger inequality). [Lean: `ABCM.wirtinger_H01`] *On the strip $Q = \{0 < \theta < \omega\}$, every $v \in H^1_0(Q)$ satisfies $(\pi/\omega)\|v\| \le \|\partial_\theta v\|$.*

Proof. For test functions, $(\partial_\theta \varphi)^2 - \nu^2 \varphi^2 = \partial_\theta(\nu \cot(\nu\theta)\varphi^2) + (\partial_\theta \varphi - \nu \cot(\nu\theta)\varphi)^2$ with $\nu = \pi/\omega$; integrate and pass to the closure. □

Lemma 6.4 (The shifted strip problem). [Lean: `ABCM.shifted_equation, ABCM.exists_strip_hessian`] *If $W \in H^1_0(Q)$ solves $(\nabla W, \nabla w) = (e^t f, w)$ and $z = e^{-t} W \in H^1_0(Q)$, then z solves a coercive, translation-invariant shifted problem; hence $\partial_t z \in H^1_0(Q)$ and $\partial^2_\theta z \in L^2(Q)$, with bounds by $\|f\|$.*

Proof. Coercivity uses the sharp Wirtinger inequality ($\pi/\omega > 1$); then Lemma 6.2 in the direction t, and the equation for the remaining derivative. □

Lemma 6.5 (Regularity at a convex corner). [Lean: `ABCM.Frame.sector_hessian`] *On a sector of opening $0 < \omega < \pi$, a solution $v \in H^1_0$ of $(\nabla v, \nabla \varphi) = (F, \varphi)$ with F vanishing outside a ball around the apex has weak second derivatives with $\|\partial_{ij} v\| \le C(\omega)\|F\|$.*

Proof. Polar coordinates $t = \log r$ map the sector conformally onto the strip Q. The transported function W lies in $H^1_0(Q)$ (conformal invariance of the Dirichlet energy and the Wirtinger inequality), and $z = e^{-t} W \in H^1_0(Q)$ by an energy estimate with truncated exponential weights. Lemma 6.4 gives H^2 regularity of z, which transfers back to the sector by the chain rule for test functions. □

Lemma 6.6 (Local geometry of convex polygons). [Lean: `ABCM.cone_classify`, `ABCM.cone_eq_sector`, `ABCM.ModelReg`, `ABCM.polygon_local`] *A nonempty finite intersection of open half-planes through* 0 *is the plane, a half-plane or a convex sector. Hence around every point of* $\overline{\Omega}$ *there are a model domain* $V \supseteq \Omega$ *and* $\delta > 0$ *with* $V \cap B(x,\delta) \subseteq \Omega$*, and every model domain satisfies the local* H^2 *estimate.*

Proof. Classification of tangent cones; the plane and half-planes satisfy the model estimate by Lemma 6.2, and convex sectors by Lemma 6.5. □

Lemma 6.7 (Localization). [Lean: `ABCM.local_hessian`] *If* $U \subseteq V$*,* $V \cap B(x_0,\delta) \subseteq U$*,* V *satisfies the model estimate at* x_0 *and the cutoff* c *is supported in* $B(x_0,\delta)$*, then* cv *has an* L^2 *Hessian for every solution* v*, bounded by* $C(\|F\| + \|v\|)$*.*

Proof. cv solves the same equation with load $cF - \sum_i (2\partial_i c\,\partial_i v + \partial_i^2 c\, v)$. □

Theorem 6.8 (H^2 regularity on convex polygons). [Lean: `ABCM.polygon_hessian`] *On a convex polygon there is* C *such that every solution* $v \in H^1_0(\Omega)$ *of* $(\nabla v, \nabla\varphi) = (F,\varphi)$ *has weak second derivatives bounded by* $C(\|F\| + \|v\|)$*.*

Proof. A smooth partition of unity near $\overline{\Omega}$ subordinate to the neighbourhoods of Lemma 6.6 reduces the estimate to Lemma 6.7. □

Chapter 7

Interpolation

Lemma 7.1 (Lagrange interpolation)**.** [Lean: `ABCM.lagInterp`, `ABCM.evalPoly_lagInterp`, `ABCM.coarseField_lagInterp`] *On a simplex with barycentric coordinates b_i and degree $k \geq 1$, the explicit Lagrange basis on the principal lattice interpolates at the lattice nodes; the interpolant on a facet depends only on the values on that facet, so cellwise interpolants of a continuous function agree on common edges.*

Proof. The basis functions are products of affine factors in the barycentric coordinates. □

Lemma 7.2 (Continuous representatives)**.** [Lean: `ABCM.exists_continuousRep`, `ABCM.continuousRep_glue`] *Under the scaled sup bound, a jet with H^2 traces on a cell has a continuous representative w on the closed cell, equal to the traces on the facets, with $|T|(w-q)(x)^2 \leq C\, E_{H^2}(J-q)$ for smooth q. Representatives on neighbouring cells agree on common edges.*

Proof. The smooth approximants converge uniformly by the sup bound (Lemma 5.10); the traces agree across edges by Lemma 5.6. □

Lemma 7.3 (Interpolation estimate on one cell)**.** [Lean: `ABCM.cell_interp_estimate`] *With a Bramble–Hilbert polynomial and a continuous representative w, the Lagrange interpolant $I_T w$ satisfies $h_T^{2i}|J - I_T w|^2_{H^i(T)} \leq C h_T^{2(k+1)}|J|^2_{H^{k+1}(T)}$ for $i \leq 2$, and the corresponding pointwise bound.*

Proof. $J - I_T w = (J-q) - I_T(w-q)$; the second term is bounded through the nodal values of $w-q$, using the sup bound and a Lebesgue constant of the Lagrange basis. □

Lemma 7.4 (Interpolation in the DG norm)**.** [Lean: `ABCM.exists_interp`, `ABCM.exists_interp_normN`] *For $J \in H^1_0(\Omega) \cap H^{p+1}(\Omega)$ realized by U there is $v_h \in V_h$, continuous across edges and zero on $\partial\Omega$, with $N_h(U - v_h) \leq C h^p |J|_{H^{p+1}(\Omega)}$ and $\|U - v_h\|_{L^2} \leq C h^{p+1}|J|_{H^{p+1}(\Omega)}$.*

Proof. Take the cellwise Lagrange interpolant of the continuous representative; the interpolants glue, so the jumps of $U - v_h$ vanish and N_h reduces to the broken H^1 and H^2 parts, bounded by Lemma 7.3. □

Chapter 8

The four stable methods

Lemma 8.1 (Liftings). [Lean: `ABCM.isLiftR_liftR, ABCM.isLiftL_liftL, ABCM.exists_liftR_upper, ABCM.exists_liftL_upper, ABCM.exists_liftR_lower`] *The liftings $r_e(\varphi)$ and $l_e(q)$ exist and are unique. Uniformly over admissible meshes: $h_e\|r_e(\varphi)\|_\Omega^2 \le C\|\varphi\|_e^2$, $h_e\|l_e(q)\|_\Omega^2 \le C\|q\|_e^2$, and $c\|[\![v_h]\!]\|_e^2 \le h_e\|r_e([\![v_h]\!])\|_\Omega^2$ for $v_h \in V_h$.*

Proof. Existence by Riesz representation in the finite-dimensional space Σ_h; the upper bounds by Cauchy–Schwarz on the edge and the discrete trace inequality; the lower bound by testing with a polynomial extension of the jump from the edge into an adjacent triangle. □

Lemma 8.2 (Bounds for the pieces of the forms). [Lean: `ABCM.exists_avgGrad_trace, ABCM.exists_jumpGrad_trace, ABCM.exists_jump_trace, ABCM.exists_fluxC_bound`] *For broken H^2 fields, uniformly over admissible meshes, $\sum_e h_e\|\{\nabla w\}\|_e^2$ and $\sum_e h_e\|[\![\nabla w]\!]\|_e^2$ are bounded by $C(G_h(w)^2 + H_h(w)^2)$, and $\sum_e h_e^{-1}\|[\![w]\!]\|_e^2 \le C\sum_K (h_K^{-2}\|w\|_K^2 + |w|_{H^1(K)}^2)$. Consequently $|C(w,v)| \le CN_h(w)N_h(v)$, and similarly for g, A_j and D_β.*

Proof. The trace inequality of Lemma 5.4 applied to the first derivatives, and Cauchy–Schwarz. □

Lemma 8.3 (Boundedness). [Lean: `ABCM.exists_std_bound`] *There is C_b, depending only on p, κ, $\eta_{\max}$ and b, with $|B(w,v)| \le C_b N_h(w)N_h(v)$ for the four stable forms and all broken H^2 fields.*

Proof. Lemma 8.2 for the edge terms and Lemma 8.1 for the lifting terms. □

Lemma 8.4 (Coercivity). [Lean: `ABCM.ip_coercive_core, ABCM.ldg_coercive_core, ABCM.exists_std_coercive`] *There is $\eta_*(p,\kappa) > 0$ such that each of the four methods, under its threshold (IP: $\eta_{\min} > \eta_*$; Bassi et al.: $\eta_{\min} > 3$; Brezzi et al. and LDG: $\eta_{\min} > 0$), satisfies $C_s N_h(v_h)^2 \le B(v_h, v_h)$ on V_h with $C_s > 0$ uniform over admissible meshes and coefficients.*

Proof. Since $(R(v), \nabla_h v) = -C(v,v)$ by the defining identity of the liftings: for IP, $C(v,v)$ is absorbed by the jump penalty when $\eta_{\min} > \eta_*$ (inverse and discrete trace inequalities); for Bassi et al. it is absorbed by A_r when $\eta_{\min} > 3$, because every triangle has three edges; for Brezzi et al. the form equals $\|\nabla_h v + R(v)\|^2 + A_r(v,v)$, and LDG contains the analogous square with A_β, so every $\eta_{\min} > 0$ suffices. The penalty controls J_h by the lower bound of Lemma 8.1, and H_h is controlled by G_h on V_h by the inverse inequality. □

Lemma 8.5 (Unique solvability). [Lean: `ABCM.existsUnique_of_injective, ABCM.eq_zero_of_normN_poly, ABCM.existsUnique_discrete`] *A bilinear form on a finite-dimensional space whose trial map is injective is uniquely solvable. N_h is a norm on V_h; hence a form that is bilinear and coercive in N_h on V_h has exactly one discrete solution for every $f \in L^2(\Omega)$.*

Proof. If $N_h(v_h) = 0$, then v_h is piecewise constant with zero jumps, and the divergence argument $\sum_K c_K^2|K| = 0$ forces $v_h = 0$. □

Lemma 8.6 (Green's formula for broken fields). [Lean: `ABCM.green_poly`] *For an exact solution u with $-\Delta u = f$ on every cell and every $v_h \in V_h$: $g(u, v_h) - C(u, v_h) = (f, v_h)_\Omega$.*

Proof. Green's formula on each cell; the edge terms combine into $C(u, v_h)$ because the traces of ∇u agree across interior edges. □

Lemma 8.7 (Consistency of the four stable methods). [Lean: `ABCM.std_consistency_core`] *If $u \in H^2(\Omega) \cap H^1_0(\Omega)$ with $-\Delta u = f$ is realized by U, then $B(U, v) = B(v, U) = (f, v)_\Omega$ for all $v \in W_h$ and each of the four stable methods.*

Proof. The jumps of U vanish, so all penalty and lifting terms containing $[\![U]\!]$ vanish, and the remaining terms reduce to Lemma 8.6; symmetry gives adjoint consistency. □

Lemma 8.8 (The abstract energy estimate). [Lean: `ABCM.energy_error`] *If B is linear in its first argument, bounded by $C_b N_h N_h$, coercive with constant C_s on V_h and consistent at U, then every discrete solution satisfies $N_h(U - u_h) \le 2(1 + C_b/C_s) N_h(U - v_h)$ for every $v_h \in V_h$.*

Proof. $C_s N_h(v_h - u_h)^2 \le B(v_h - u_h, v_h - u_h) = B(v_h - U, v_h - u_h) \le C_b N_h(U - v_h) N_h(v_h - u_h)$ and the triangle inequality. □

Lemma 8.9 (The duality argument). [Lean: `ABCM.l2_duality`] *Let B be bilinear, bounded on broken H^2 fields, consistent at U and adjoint consistent on W_h; assume H^2 regularity with constant C_{reg} and an $O(h)$ interpolant for H^2 jets. Then $\|U - u_h\|_{L^2} \le C_b C_I C_{reg}\, h\, N_h(U - u_h)$.*

Proof. Aubin–Nitsche: test the adjoint problem with the error, subtract the interpolant of the dual solution using Galerkin orthogonality, and bound with C_b. □

Chapter 9

Flux formulations

Lemma 9.1 (Elimination of the auxiliary variable)**.** [Lean: `ABCM.flux_first, ABCM.flux_second, ABCM.flux_vdb_aux`] *For each method and each triangle,* $-(u_h, \operatorname{div}\tau)_K + \langle \hat{u}, \tau \cdot n_K\rangle_{\partial K} = (\sigma_h^*, \tau)_K$*, where* $\sigma_h^* \in \Sigma_h$ *is the eliminated auxiliary field; and* $(\sigma_h^*, \nabla w) - \sum_K \langle \hat{\sigma} \cdot n_K, w\rangle_{\partial K} = B_h(u_h, w)$ *for all* $w \in V_h$*.*

Proof. Integration by parts on each triangle and the defining identities of the liftings, which turn the edge terms of $\hat{u}$ into R, L_β or A_β; the edge terms of $\hat{\sigma}$ give C, the penalties and the lifting products. □

Chapter 10

NIPG, pure penalties and superpenalties

Lemma 10.1 (Stability of the penalty methods). [Lean: `ABCM.exists_penalty_bound`, `ABCM.exists_penalty_coercive`, `ABCM.super_coercive`] *NIPG and the pure penalty forms are bounded on W_h and coercive on V_h in N_h, for every $\eta_{\min} > 0$; the superpenalized forms are coercive in their superpenalty norms.*

Proof. For NIPG the terms $\mp C$ cancel in $B(v, v)$; the pure penalties contain no C. Boundedness as in Lemma 8.3. □

Lemma 10.2 (Broken Poincaré inequality). [Lean: `ABCM.exists_broken_poincare`] *On a convex polygon there is C_P with $\|v\|_{L^2(\Omega)} \le C_P N_h(v)$ for all $v \in W_h$, uniformly over admissible meshes.*

Proof. Duality: solve $-\Delta\psi = v$ with H^2 regularity and integrate by parts cellwise against v. □

Lemma 10.3 (Jump control by the superpenalty). [Lean: `ABCM.jumpSq_le_superJ`, `ABCM.jumpSq_le_superR`] *$J_h(v)^2 \le h^{2p}\eta_{\min}^{-1} A_j^{\sup}(v, v)$, and the analogous bound with $A_r^{\sup}$.*

Proof. $h_e \le h$ in the weights; for $A_r^{\sup}$ the lower bound of Lemma 8.1. □

Lemma 10.4 (Error estimates with consistency defects). [Lean: `ABCM.super_energy`, `ABCM.super_l2`, `ABCM.super_assembly`] *For a coercive bounded form whose primal defect is $(1-\varepsilon)C(u, \cdot)$, bounded by Γ in the method's norm, every discrete solution satisfies $N(U-u_h) \le 2\big((1+C_b/C_s)N(U-v_h) + |1-\varepsilon|\Gamma/C_s\big)$; a duality version gives the L^2 estimate when the adjoint defect is $(1+\varepsilon)C(\psi, \cdot)$. For the three superpenalized methods these give the h^p and h^{p+1} estimates.*

Proof. As in Lemmas 8.8 and 8.9, carrying the defect; by Lemma 10.3 the defect is of order h^p. □

Chapter 11

The Baumann–Oden method

Lemma 11.1 (Stability and solvability for $p \geq 2$). [Lean: `ABCM.bo_formBO_self, ABCM.bo_exists_meanRemoval, ABCM.bo_existsUnique`] *$B_{BO}(v,v) = G_h(v)^2$: the skew edge terms cancel. On V_h, $N_h(v_h - P_0 v_h)^2 \leq C G_h(v_h)^2$, where P_0 is the elementwise mean. For $p \geq 2$ the Baumann–Oden problem is uniquely solvable.*

Proof. The jumps of piecewise constants are controlled by testing with the quadratics $\chi_{T,j} = |x - P_j|^2/(4|T|)$, whose normal-derivative moments on the edges are δ_{jl}; this needs $p \geq 2$. Injectivity then follows as in Lemma 8.5. □

Lemma 11.2 (Failure for $p = 1$). [Lean: `ABCM.oneTri_admissible, ABCM.bo_p1_counterexample`] *The mesh consisting of the triangle with vertices $(0,0)$, $(1,0)$, $(0,1)$ is admissible, and for $p = 1$ the load $f = 1$ has no Baumann–Oden solution on it.*

Proof. For affine u_h, $B_{BO}(u_h, 1) = 0$, while $(1,1)_\Omega = |\Omega| > 0$. □

Chapter 12

The original Bassi–Rebay method

Lemma 12.1 (The Bassi–Rebay form is a square)**.** [Lean: `ABCM.BR.formBR_poly, ABCM.BR.discreteGrad_green`] *On $V_h \times V_h$, $B_{BR}(w,v) = (\mathcal{D}_h w, \mathcal{D}_h v)_\Omega$. For $v \in V_h$ and $\tau \in \Sigma_h$: $(\mathcal{D}_h v, \tau) = -(v, \operatorname{div}_h \tau) + \sum_{e \text{ interior}} \langle \{v\}, [\![\tau]\!] \rangle_e$.*

Proof. Expand $(\nabla_h w + R(w), \nabla_h v + R(v))$ with the defining identity of the liftings; the Green identity follows from integration by parts on each triangle. □

Lemma 12.2 (Local Raviart–Thomas constructions)**.** [Lean: `ABCM.BR.exists_ref_bubble, ABCM.BR.exists_local_lift`] *On every triangle there are polynomial vector fields of degree at most p with prescribed compatible normal traces on the edges and prescribed divergence of degree at most $p-1$, with bounds uniform under shape regularity.*

Proof. Lowest-order Raviart–Thomas fields for the normal traces, edge bubbles $\lambda_a \lambda_b \sigma (P_b - P_a)$ for the divergence, transported from a reference triangle. □

Lemma 12.3 (The kernel)**.** [Lean: `ABCM.BR.brKernel_iff`] *For $v_h \in V_h$: $\mathcal{D}_h v_h = 0$ if and only if $P_{p-1} v_h = 0$ on every triangle and $\{v_h\} = 0$ on every interior edge.*

Proof. Test $(\mathcal{D}_h v_h, \tau) = 0$ (Lemma 12.1) with fields supported on one triangle: edge bubbles force $P_{p-1} v_h = 0$, and fields with prescribed normal traces force $\{v_h\} = 0$. □

Lemma 12.4 (Solvability)**.** [Lean: `ABCM.BR.brSolvable_iff, ABCM.BR.isDiscreteSolution_iff_kernel, ABCM.BR.discreteGrad_unique`] *The problem is solvable if and only if $f \perp Z_h$; the solutions form one coset of Z_h, and $\mathcal{D}_h u_h$ and $P_{p-1} u_h$ do not depend on the solution.*

Proof. A finite-dimensional Fredholm alternative for the symmetric form $(\mathcal{D}_h \cdot, \mathcal{D}_h \cdot)$, whose kernel is Z_h; P_{p-1} vanishes on Z_h by Lemma 12.3. □

Lemma 12.5 (The conservative interpolant of the flux)**.** [Lean: `ABCM.BR.exists_consInterp`] *Let $u \in H^2(\Omega)$ solve $-\Delta u = f$ with f elementwise polynomial of degree at most $p-1$. For every $w \in V_h$ there is $\sigma \in \Sigma_h$ with $(\mathcal{D}_h v_h, \sigma) = (f, v_h)$ for all $v_h \in V_h$ and $\|\nabla u - \sigma\|^2 \le C\big(G_h(u-w)^2 + H_h(u-w)^2\big)$.*

Proof. Build σ triangle by triangle from ∇w with Lemma 12.2, with single-valued normal traces and $\operatorname{div} \sigma = -f$. □

Lemma 12.6 (Error estimates)**.** [Lean: `ABCM.BR.exists_flux_error, ABCM.BR.exists_proj_error`] *For elementwise $\mathbb{P}_{p-1}$ loads and $u \in H^{p+1}(\Omega)$, every discrete solution satisfies $\|\nabla u - \mathcal{D}_h u_h\|^2 \le C h^{2p} |u|^2_{H^{p+1}}$ and $\|P_{p-1}(u - u_h)\|^2 \le C h^{2(p+1)} |u|^2_{H^{p+1}}$.*

Proof. Compare with the conservative interpolant of Lemma 12.5 and the interpolant of Lemma 7.4; the projected estimate by duality with the H^2 regularity of Theorem 4.1. □

Lemma 12.7 (The full L^2 error is not controlled)**.** [Lean: `ABCM.BR.oneTri_br_example`] *For every $p \geq 1$, on the one-triangle mesh, $z = x_0^p - P_{p-1}(x_0^p)$ is a nonzero element of Z_h; every tz solves the problem with $f = 0$, and $(z, z) \neq 0$, so $f = z$ has no solution.*

Proof. There are no interior edges, so by Lemma 12.3 the kernel consists of the polynomials of degree at most p orthogonal to $\mathbb{P}_{p-1}$. □

Chapter 13

Correspondence with the paper

- **Formalized:** the principal quantitative results for all nine methods of Table 6.1 of the paper and the three superpenalized methods: the flux–primal equivalence (Section 3.2), consistency and adjoint consistency (Section 3.3), boundedness and stability (Section 4), the error estimates of Section 5.1 for the four stable methods, NIPG, the pure penalties and the superpenalties (Section 5.2), Baumann–Oden (Section 5.3.1) and the original Bassi–Rebay method (Section 5.3.2).
- **Setting:** Ω is a bounded open convex polygon; meshes are conforming shape-regular triangulations, without quasi-uniformity. The H^2 regularity of the Dirichlet problem, used by the paper's duality arguments, is proved (Theorem 4.1).
- **Norms:** the DG norm N_h includes the scaled broken H^2 term, which makes it uniformly equivalent to the lifting-based norm (4.2) of the paper on V_h.
- **Qualifications made explicit:** Baumann–Oden is analysed for $p \geq 2$, with a counterexample for $p = 1$; for Bassi–Rebay the solution is unique only modulo Z_h, the estimates concern $\mathcal{D}_h u_h$ and $P_{p-1} u_h$, and the full L^2 error is shown to be uncontrolled.
- **Not formalized:** the historical discussion, the qualitative comparisons and the open questions of the paper.